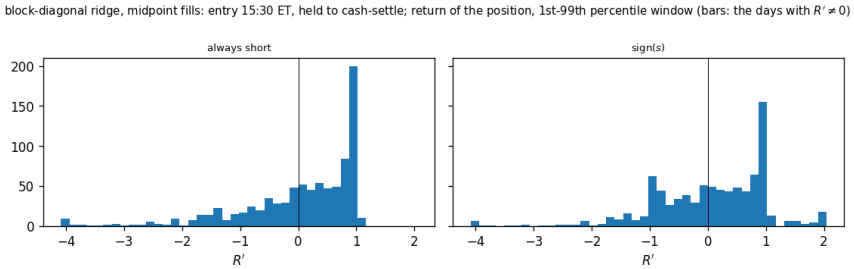
0DTE nearest-OTM straddle: per-rule return summary, entry 15:30 ET, held to cash-settle, models compared on the same 866 days

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.017	1.115	-9.38	1.08	-2.26	9.18	0.44	0.24	0	0.0	-0.23
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.027	1.115	-9.38	5.49	-1.47	9.19	0.71	0.39	148	17.1	-0.09
block-diagonal ridge	866	0.081	1.113	-6.37	9.38	0.05	9.12	2.14	1.15	234	27.0	0.68
block-diagonal ridge, without the FOMC columns	866	0.082	1.113	-6.37	9.38	0.04	9.12	2.17	1.17	236	27.3	0.70
LightGBM†	866	0.103	1.111	-6.37	9.38	0.02	9.19	2.74	1.48	165	19.1	1.01
XGBoost†	866	0.091	1.112	-9.38	5.49	-1.34	9.60	2.40	1.30	162	18.7	0.82
lasso (causally tuned)†	866	0.102	1.111	-6.37	9.38	-0.15	9.25	2.71	1.46	184	21.2	0.99
lasso ($\alpha = 10^{-4}$)	866	0.064	1.114	-6.37	9.38	-0.22	9.15	1.70	0.91	246	28.4	0.44
elastic net (causally tuned)†	866	0.062	1.114	-6.37	9.38	-0.11	9.12	1.65	0.89	182	21.0	0.42



Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

One trade a day: enter the nearest-OTM straddle at 15:30 ET, hold those strikes to the official close, and delta-hedge every 30 minutes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule at the crossed spread: entry at the touch (the ask when long, the bid when short) and cash-settle at the official close (no exit spread). **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one remaining-session short per expiration day (enter at this clock, hold those strikes to the close, delta-hedge every 30 minutes). It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's thirteen overlapping remaining-session books. That reserved name is the book's daily P&L after pooling the session. The thirteen books overlap; they are not thirteen successive holds. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the thirteen positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat. At the 09:35 entry the forecast is the one issued at 09:30 for the 09:30–10:00 bar, w_t is that bar's share, and h_t is the chain's hours to the close.